# Over-Pledging in Crowdfunding: Evidence from Kickstarter

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In the context of online reward-based crowdfunding, it is a recurring phenomenon for backers to pledge financial amounts exceeding the established requisites. This paper delineates the scope, variations, and potential correlates of such *over-pledging* behavior, utilizing a pledge-level dataset sourced from Kickstarter. First, over-pledged funding was common (22% of backers), substantial (19% of the total pledge amount), and essential to the success of projects. Second, significant variations in over-pledging across projects were observed. On average, the proportion of over-pledged funding to the total pledged amount for a project within dance/theatre categories surpassed that of a project within games/design categories by more than 20%. In examining the over-pledging backers, we identified two types: *donors*, who contribute without seeking a reward, and *funders*, who pledge additional sums for rewards. In alignment with extant literature on voluntary contributions, our empirical evidence lends support to (pure or impure) altruism and various strategic considerations as potential drivers of donors' contributions. Conversely, by contrasting the over-pledging patterns of funders with those of donors, we pinpoint the equitable distribution of surplus as a significant motivation underlying funders' over-pledging. Leveraging an exogenous shock at the project level, we further ascertain that funders' perceived fair value tends to escalate in response to successful crowdfunding initiatives. Overall, our findings underscore the significance of over-pledging and underline the necessity for creators to customize strategies for project setup according to their target audience, distinguishing between funders and donors.

*Key words*: Crowdfunding; Over-pledging; Pay-what-you-want; Altruism; Inequity aversion

## 1. Introduction

Crowdfunding has evolved into a preferred fundraising method for startups and smaller entities embarking on new ventures (Barnett 2015). One notable form of crowdfunding is reward-based crowdfunding, represented by prominent online platforms like Kickstarter and Indiegogo. In this model, individuals, often known as *backers*, pledge financial support to projects proposed by *creators* and, in turn, receive rewards. Reward-based crowdfunding projects typically provide an array of reward tiers, each linked to a specific price. By contributing an amount equal to a designated price, a backer can obtain a particular reward. Uniquely, these online reward-based crowdfunding platforms offer backers the opportunity to pledge sums exceeding the required amount, a phenomenon referred to as *over-pledging*. This paper aims to delve into this over-pledging behavior. We aim to determine its prevalence in online crowdfunding and assess its significance to the overall success of crowdfunding initiatives. Further, we seek to understand the potential motivations driving backers' over-pledging actions and identify which factors - associated with the creators, the projects, or the platform itself - correlate with the intensity of over-pledging.

We address these questions with descriptive evidence from a unique data set from Kickstarter, the market leader of online reward-based crowdfunding platforms. Kickstarter adopts an all-or-nothing mechanism, where a project will be successfully funded if and only if the total pledge amount exceeds a prespecified funding target. The data set consists of all projects launched between Aug 1, 2016, and Dec 31, 2016. The data includes project characteristics (e.g., project category, funding target, reward levels, and campaign duration), creator characteristics (e.g., geographical location, past experience in backing and launching projects on the platform), pledge-level records (timing and amount of every pledge, the attained reward level and shipping cost, and over-pledging amount), and click-through traffic (hourly) from different online channels.

Our data provides empirical evidence demonstrating that over-pledging behavior is notably widespread in online crowdfunding. On average, it was observed that 22% of the backers associated with a given project contributed in excess of the required amount for their selected rewards. This over-pledged funding represented a significant 19% of the total pledge amount garnered by each project. Such surplus funding manifested as a critical determinant in the successful attainment of project funding objectives. In the (hypothetical) absence of these additional contributions, the overall rate of successful fundraising would have plummeted. More specifically, given all else the same, approximately 46% of the projects that were successfully funded in our dataset would have unfortunately failed to meet their funding targets.

To measure the *over-pledging intensity* of a crowdfunding project, we construct the *over-pledging score* (OP score, thereafter) as the ratio between the extra payment and the total pledge amount of the same project. Ranking the projects in the sample according to the OP scores generates a *over-pledging spectrum* with significant variations ranging from 0 to 1. In an effort to elucidate these variations, we compare projects' OP scores, categorized by both project and creator characteristics. Our findings indicate a significantly heightened intensity of over-pledging in projects that offered products within the fine arts domain (including dance, theatre, journalism, film & video, and music). Additionally, this intensity was markedly elevated when project creators were based in the United States, as opposed to other countries, or when the creators possessed no prior experience with Kickstarter crowdfunding. These results suggest intriguing trends related to the geographical location, field of endeavor, and prior crowdfunding exposure of creators, all of which seemingly influence the magnitude of over-pledging.

Within the purview of over-pledging, our research identifies two distinct types of backers - those we categorize as *donors* and *funders*. A donor refers to a backer who makes a financial contribution *without* seeking a product in return, whereas a funder refers to a backer who pledges an amount exceeding the required price for a particular reward level, while still receiving a product. Drawing from the existing literature on voluntary contributions in the contexts of philanthropic donations and pay-what-you-want (PWYW) (Fehr and Schmidt 2006, Natter and Kaufmann 2015), we propose that donors could be driven by a spectrum of motivations, encompassing both pure and impure forms of altruism, along with other strategic considerations. Such considerations may include the desire to ensure a project's success, a sense of reciprocity, among others. These motivations are often influenced by the perceived need of a given project. Indicators of need, such as a substantial gap between the pledged amount and the funding goal, or the project creator's lack of prior experience, might engender empathetic or sympathetic responses from backers, which in turn, could result in amplified over-pledging. Our data, consistent with this proposition, suggests a heightened intensity of over-pledging from donors for larger-scale projects, projects helmed by inexperienced creators, and projects managed by creators based in the United States.

These factors may also impact the over-pledging propensities among funders. Nonetheless, a crucial distinction exists between donors and funders, characterized by the exchange of a product or service exclusively in the latter's case. This implies that the funders' perceived value of the product or service could influence their over-pledging tendencies. In the context of PWYW, plenty of empirical evidences suggest that consumers often demonstrate a preference for a balanced distribution of surplus between themselves and sellers (Kim et al. 2009, Gneezy et al. 2012, Schmidt

et al. 2014). Our empirical investigation strongly supports the notion that a fair distribution of surplus plays a substantial role in determining the intensity of over-pledging among funders. This is reflected in the divergent influences of creator and project characteristics on the over-pledging behaviors of donors versus funders. Generally, the perceived surplus value tends to be higher for smaller-scale projects, projects that eschew video content, projects managed by experienced and/or non-U.S.-based creators, and projects typified by regular updates.

To fortify our findings, we exploit two exogenous shocks at both project and platform levels. We demonstrate that funders' perceived value tends to increase consequent to successful crowdfunding endeavors. However, the perceived fair value of the project remains relatively consistent between customers drawn from social media and those from other sources. Our analysis underscores the dynamic nature of perceived value and its impact on over-pledging behavior in crowdfunding scenarios.

**Related Literature**

Our study is closely related to two streams of literature: reward-based crowdfunding and voluntary contributions in the contexts of philanthropic donations and pay-what-you-want (PWYW). In recent years, online crowdfunding has become a popular avenue for entrepreneurs to raise capital in the early stage (Wilson and Testoni 2014, Allon and Babich 2020). Our paper focuses on reward-based non-equity crowdfunding (Belleflamme et al. 2015). The recent success of platforms, such as Kickstarter and Indiegogo, has attracted a significant number of researchers, on both the analytical and empirical sides, to study this industry. Most of the analytical research focuses on the optimal project decisions, including products (rewards), prices, communication, funding targets, and campaign duration. For instance, Hu et al. (2015) study product and pricing decisions and demonstrate how the crowd aspect of this business model can make it more appealing to offer multiple products. Chakraborty and Swinney (2020) examine the role of the funding target in signaling product quality. Liu et al. (2021) investigate the optimal design of the product menu and the funding target from the entrepreneur's perspective. Du et al. (2022) consider entrepreneurs' optimal dynamic strategy in updating projects throughout a crowdfunding campaign. Although our paper is empirical and focuses on the backers' over-pledging behavior, our results link the over-pledging behavior to crowdfunding campaign decisions studied in this literature.

The unique features of the reward-based crowdfunding model have attracted empirical researchers from a broad range of disciplines. The existing empirical literature has studied the factors contributing to the success of projects (Mollick 2014, Li et al. 2019, Candoğan et al. 2024), the U-shape funding pattern over the duration of crowdfunding campaigns (Kuppuswamy and

Bayus 2018), and the goal gradient effect under the all-or-nothing crowdfunding mechanism (Kuppuswamy and Bayus 2017, Dai and Zhang 2019). Some papers have also examined the motives of the backers. For instance, Agrawal et al. (2015) show that the backers located geographically closer to the creators were less responsive to the cumulative funds already raised. They propose a plausible explanation for this phenomenon that the local backers could be more likely to have an offline social relationship (e.g., friends and family members) with the creators. Zvilichovsky et al. (2018) find evidence of reciprocity where the success rate of a creator is positively correlated with the creator's past experience of backing other projects. We contribute to the empirical literature with descriptive evidence offering a detailed view of backers, especially their over-pledging behavior.

The paper closest to ours is Deb et al. (2024), which also considers voluntary payments from the backers on Kickstarter. They establish that voluntary payments from the backers represent a significant portion of the total pledge amount in crowdfunding. However, our study markedly diverges from Deb et al. (2024) with entirely distinct research objectives. Their goal is to examine the role of prosocial backers in contribution dynamics by understanding the strategic role of over-pledging backers in changing the behavior of regular backers. On the contrary, our primary goal is to understand the variations in over-pledging intensity and creator-, project- and platform-level correlates with over-pledging behaviors. Moreover, unlike the high-frequency data extracted from the Kickstarter website in Deb et al. (2024), we obtained pledge-level data from the company's internal records, which allows us to separate donors from funders and accurately measure funders' over-pledging.

On the other hand, our paper is related to research on voluntary contributions in the contexts of philanthropic donations and pay-what-you-want (PWYW), where consumers can decide the price they pay for the goods or service they procure. Incentives for voluntary contributions typically include: altruism, fairness and other strategic considerations (Fehr and Schmidt 2006, Natter and Kaufmann 2015). For instance, within the category of altruistic-oriented motives, one such motivator can be pure altruism, wherein individuals contribute out of a genuine concern for the well-being of others (Piliavin and Charng 1990, Small and Cryder 2016). Alternatively, contributors may be influenced by impure altruism, marked by elements such as deriving moral satisfaction/warm glow (Andreoni 1990, Jung et al. 2017). In addition to altruism, an abundant body of empirical evidence substantiates the notion that consumers frequently experience an obligation to reciprocate equitably towards sellers by providing remuneration exceeding the minimum required within the context of PWYW (Kim et al. 2009, Gneezy et al. 2012, Schmidt et al. 2014). This notion of fairness is commonly referred to as inequity aversion (Fehr and Schmidt 1999). Lastly, strategic

factors such as reciprocity (Rabin 1993, Fehr and Schmidt 2006, Regner and Barria 2009), the motivation to keep seller in the market (Schmidt et al. 2014) and etc. can also be the motives for voluntary contributions. From the fundraisers' perspective, McCardle et al. (2009) and Castro and Rodilitz (2024) examine the optimal design of donation tier levels and suggested donation amounts, respectively, to incentivize donor contributions.

The reward-based crowdfunding can be understood as a particular instance of the PWYW model, wherein a project creator sets a prerequisite price for each reward level. This is equivalent to PWYW with a minimum price. Thus, our work augments the academic discourse on voluntary contributions within PWYW, providing comprehensive empirical evidence drawn from the operations of a crowdfunding platform. More interestingly, we identify two different types of backers who over-pledge, i.e., funders and donors. Both these types could be spurred to over-pledge due to altruism and/or other strategic considersations such as reciprocity. However, there exists a fundamental divergence, in that the contributions of donors do not result in any product or service exchanges. Consequently, the perceived fair value of products or service - a key determinant of voluntary contributions (Kim et al. 2009, Gneezy et al. 2012, Schmidt et al. 2014) - does not exert influence on donors' propensity to over-pledge. By performing a difference-in-difference analysis between funders and donors, we can disentangle fairness from other motivations underlying voluntary contributions, a feat usually unattainable in other scenarios, and thereby comprehend how it may be influenced by creator-, project- and platform-level factors.

## 2. Data and Over-Pledging Patterns in Crowdfunding

### 2.1. Background and Data

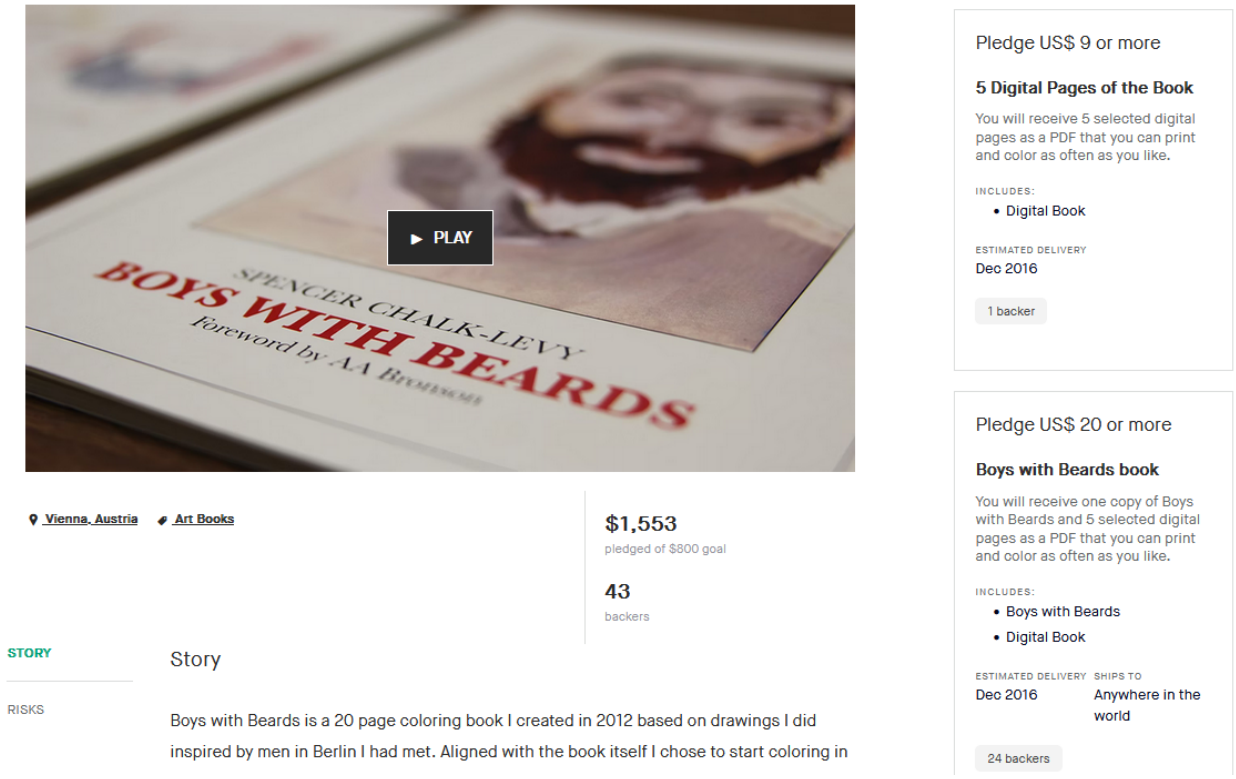
Reward-based crowdfunding, as a form of entrepreneur financing, has grown rapidly in recent years. We obtained the data set from Kickstarter, the leading online reward-based crowdfunding platform. Since its inception in Apr 2009, around 269,000 projects have been successfully funded on Kickstarter, with a total pledge amount at around $8.41 billion (as of Nov 2024).[1] Kickstarter adopts an all-or-nothing mechanism: a project creator needs to specify a monetary target at the launch of the project, and the project will be successfully funded if and only if the total pledge amount from the backers exceeds the monetary target by the end of the crowdfunding campaign. Otherwise, the project fails, and the funds will be returned to the backers.

Kickstarter allows a creator to specify multiple levels of rewards at the beginning of a crowdfunding campaign. Each reward level has a price, which is the minimum amount a backer must

[1] https://www.kickstarter.com/help/stats

pay for the reward. Reward levels can be seen as quality levels; for instance, a writer can offer a digital version and a physical version of the same book as different reward levels (See an example in Figure 1). A backer can secure a reward by pledging the amount equal to the price specified for the reward. Kickstarter allows backers to pledge more than the specified price without receiving any extra tangible benefit. A backer can also pledge money without asking for any reward. Any such extra payment beyond the required amount is referred to as *over-pledging* in this paper.

**Figure 1 A Sample Project from Kickstarter**




*Note.* Link of the project: https://www.kickstarter.com/projects/spencerchalklevy/boys-with-beards-a-coloring-book-for-all-ages.

The initial data set consists of 18,411 projects launched in Kickstarter from Aug 1, 2016, to Dec 31, 2016 (152 days in total).[2] For each project, the data contains project characteristics, creator characteristics, and pledge dynamics. First, project characteristics include the funding target, reward levels with corresponding prices, campaign duration, and the product category. Kickstarter classifies all crowdfunding campaigns into 15 categories, including dance, theater, journalism, film

[2] The observation window of the data set goes beyond Dec 31, 2016, as many crowdfunding campaigns ended after Dec 31, 2016.

& video, music, food, art, publishing, crafts, technology, fashion, photography, comics, games, and design. Second, creator characteristics include the creator's geographical location and past experience in launching or backing any projects. Finally, the data contains the most detailed information on every pledge, including the time when the pledge was made, the amount of pledge, the shipping cost, and the attained reward. Note that our transaction-level data facilitates us to accurately calculate over-pledging, which equals to the pledged amount minus the required price for a reward and the shipping cost for each specific shipping destination.

We filtered out cases from our analysis as follows. First, Kickstarter classifies all projects into four types according to the outcomes, namely "successful", "failed", "suspended", and "canceled". A project may be suspended if it violates rules set by Kickstarter. Project creators can voluntarily cancel their projects at any point in time during a crowdfunding campaign. We consider these two types of projects to be irregular and therefore excluded them from the analysis. Second, we exclude any project with an extreme funding target, i.e., lower than \$20 (the 0.5th percentile of the distribution of funding targets) or higher than \$500,000 (the 99.5th percentile). Third, we exclude all the projects with an extremely short duration, i.e., shorter than or equal to 2 days (48 hours). A typical crowdfunding campaign on Kickstarter lasts for around 30 days (or 720 hours, as shown in Table 1). Backers may not visit the crowdfunding platform frequently, and a very short duration would not give the creator the chance to reach enough potential backers. Lastly, we exclude those projects with fewer than 5 backers in the analysis. As a result of this sample selection process, we have a total of 11,219 projects for the analysis. The filter procedures and the number of projects filtered-out by each step is summarised in the e-companion.

Table 1 presents the summary statistics for those selected 11,219 projects. The average funding target was \$15,826.72, with a median of \$6,000, indicating that the distribution of funding targets is heavily skewed with a long right tail. An average crowdfunding campaign ran for 770.77 hours (around 32 days). Campaign creators chose, on average, 9.68 reward levels for each project. However, as shown in Table 1, there was a significant variation in the reward levels, ranging from 1 to 179. Around 71% of projects included at least one video on their project webpage. Videos were not required to launch a project. However, Kickstarter advises creators that projects with good videos are much more likely to succeed.

A project can have multiple creators, including one lead creator designated by Kickstarter and the rest of the creator team labeled as collaborators. That being said, more than half of the projects were launched by a single creator, as reflected by the median "number of collaborators" being 0. The demographic information mainly concerns the lead creator of a project. Among all of

**Table 1 Summary Statistics**

| | Mean | Std. Dev. | Median | Minimum | Maximum |
|---|---|---|---|---|---|
| **Project Characteristics** | | | | | |
| Funding target ($) | 15,826.72 | 32,609.68 | 6,000 | 20 | 497,806 |
| Duration (hours) | 770.77 | 246.17 | 720 | 72 | 2,093 |
| Number of reward levels | 9.68 | 6.35 | 8 | 1 | 179 |
| Video (= 1 if yes; = 0 otherwise) | 0.71 | 0.45 | 1 | 0 | 1 |
| **Creator Characteristics** | | | | | |
| First-time creator (= 1 if yes; = 0 otherwise) | 0.73 | 0.45 | 1 | 0 | 1 |
| Number of projects backed before | 6.75 | 24.92 | 0 | 0 | 749 |
| Number of collaborators | 0.45 | 0.94 | 0 | 0 | 13 |
| US (= 1 if yes; = 0 otherwise) | 0.66 | 0.47 | 1 | 0 | 1 |
| **Project Outcome** | | | | | |
| Total number of backers | 246.35 | 1,940.37 | 52 | 6 | 158,514 |
| Total pledge amount ($) | 20,412.69 | 146,639.57 | 3,185 | 14 | 9,420,134 |
| Success (= 1 if yes; = 0 otherwise) | 0.62 | 0.49 | 1 | 0 | 1 |

the projects studied, 73% of project creators were entrepreneurs launching their first Kickstarter project. On average, a project creator had backed 6.75 projects before launching her/his own crowdfunding campaign under study; however, more than half of them had no experience in backing other projects. Finally, 66% of project creators were based in the US.

In terms of project outcome, on average, a project raised $20,412.69 from around 246 backers. In our sample, 6,962 projects were successfully funded at the end of their crowdfunding campaign, leading to a success rate of 62.06%, which is higher than the statistics published by Kickstarter.[3] This is due to our sample selection procedure which excluded all the projects with fewer than 5 backers.

### 2.2. Backers' Over-Pledging Patterns on Kickstarter

Unlike conventional business models where consumers pay the listed prices for products or services, the backers can choose to pledge more than required in crowdfunding. Specifically, a backer who over-pledges may request a reward by paying more than required or choose to pledge without a reward. Later we will distinguish between these two types of over-pledging and compare them from various perspectives. In this section, we first present descriptive statistics to establish that backers who over-pledge are common, and over-pledged funding comprises a substantial proportion of the total funding.

[3] The all-time success rate of projects on Kickstarter was around 41.87%, as of Nov 2024 (source: https://www.kickstarter.com/help/stats).

**Table 2 Over-Pledging from Backers**

| | Mean | Std. Dev. | Median | Minimum | Maximum |
|---|---|---|---|---|---|
| Over-pledging ($) | 1,694.88 | 10,009.95 | 386 | 0 | 843,433 |
| Over-pledging/Total pledge amount | 0.19 | 0.21 | 0.11 | 0 | 1 |
| Number of OP backers | 24.91 | 94.70 | 9 | 0 | 4,047 |
| Proportion of OP backers | 0.22 | 0.18 | 0.17 | 0 | 1 |

*Note*: OP backers refer to those backers who over-pledged; Proportion of OP backers = the number of over-pledging backers/the total number of backers.

Table 2 summarizes the number of backers who over-pledged and the level of over-pledging across all the projects in the sample. On average, a project received $1,694.88 of over-pledging from 24.91 backers, and the over-pledged funding accounted for around 19% of the total pledge amount for a project. Note that the ratio between the amount of over-pledging and the total pledge amount could vary from 0 to 1, where 0 implies no over-pledged backer and 1 means that all the pledge amount comes from the over-pledging without a reward. The average ratio between the number of over-pledged backers and the total number of backers is 0.22.

**Table 3 Impact of Over-pledged Funding on the Success of Projects**

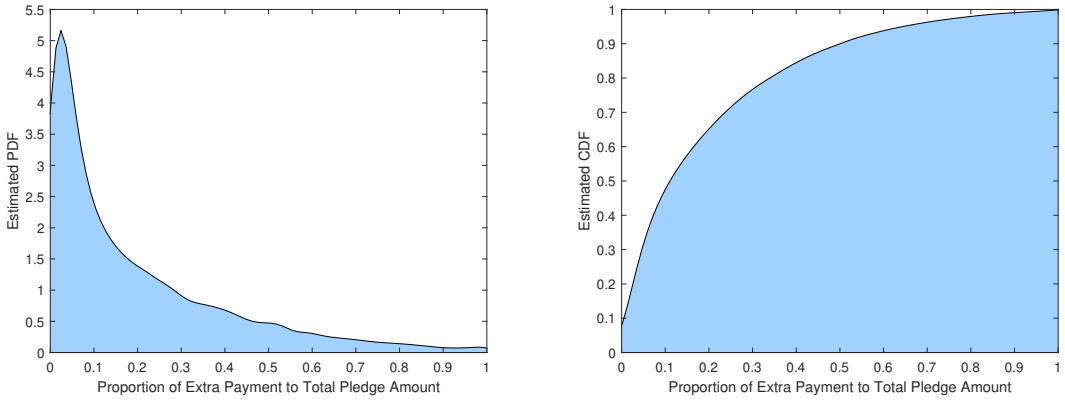
| | With Over-pledging (Sample) | Without Over-pledging (Hypothetical) |
|---|---|---|
| Number of successful projects | 6,962 | 3,766 |
| Success rate (%) | 62.06 | 33.57 |

To illuminate the significant impact of over-pledging on project success, we conduct a simple counterfactual exercise in a hypothetical scenario without any extra payment. To obtain an estimate for this hypothetical scenario, we deduct the extra payment from the total pledge amount for each project and investigate how the project's success rates could be affected. As shown in Table 3, all else being equal, the number of successful projects would drop from 6,962 to 3,766 after taking out all over-pledged funding, a 46% reduction in the number of successful projects, leading to a much lower success rate of 33.57%. The magnitude of the effect is substantial, and the economic impact would be significant to both project creators and crowdfunding platforms. Since this exercise does not take into account that earlier pledges have positive externalities on later arrivals (Kim et al. 2020), the actual impact of over-pledging could be even more significant than what is summarized in Table 3.

Next, we further explore the heterogeneity of over-pledging patterns in crowdfunding. We define a project's *over-pledging score* (OP score) as the ratio between the total extra payment and the

total amount the project receives. The OP score measures the intensity of over-pledging, i.e., the extent to which backers are willing to contribute for no tangible benefits to support the project. We use the *over-pledging spectrum* (OP spectrum) to describe the distribution of OP scores associated with the crowdfunding projects. The spectrum ranges from 0 to 1. An OP score of 0 corresponds to the traditional retail model where buyers pay the listed prices; in contrast, an OP score of 1 corresponds to the charity donation model.

**Figure 2 Over-pledging Spectrum of Crowdfunding Projects**




(a) Probability Density Function (b) Cumulative Distribution Function

The OP spectrum is constructed using the smoothing kernel density estimation (KDE), with a normal kernel function evaluated at 100 equally spaced points covering the range of OP scores. The results are shown in Figure 2. Figures 2(a) and 2(b) present the probability density function (PDF) and the cumulative distribution function (CDF) of the OP spectrum, respectively. First, the figures show that, somewhat surprisingly, the spectrum covers the entire range between 0 and 1. Second, the spectrum is right-skewed with a fat right tail. Two-thirds of projects have an OP score of less than 0.21, while at the other end of the spectrum, around 10% of projects receive more over-pledged funds than the required payments for rewards. The substantial heterogeneity in OP scores across the crowdfunding projects underscores the importance of further understanding the drivers for over-pledging behavior.

*Over-pledging spectrum by project outcome.* Table 4 compares over-pledging between the projects that succeeded and failed. On average, successful projects received a greater amount of extra payment from a larger number of over-pledged backers than the failed projects. This might be expected because, on average, successful projects received a greater amount of pledges from a larger number of backers. However, surprisingly, the average amount of over-pledging in proportion to

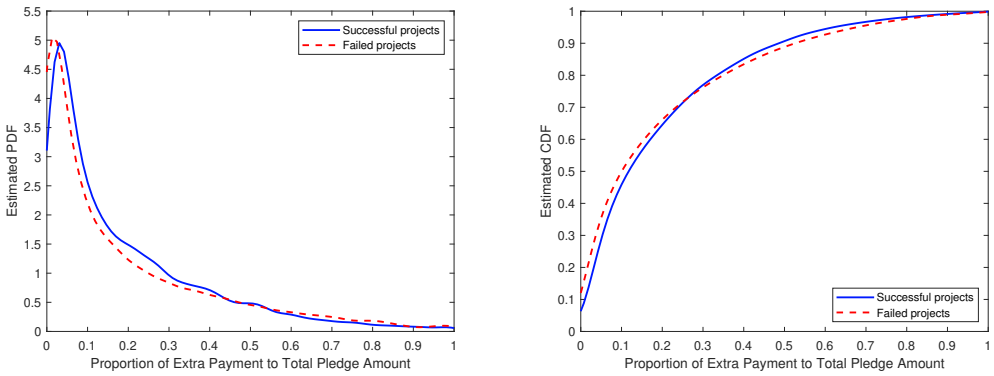
the total pledge amount was nearly the same (around 0.19) for the successful and failed projects. Similarly, the average proportion of backers who are willing to over-pledge was close, with 0.24 among failed projects, and 0.21 among successful projects. Overall, the intensity of over-pledging did not differ much by project success.

**Table 4 Comparison of Over-pledging Between Successful and Failed Projects**

| | Successful Projects | | | | | Failed Projects | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | Mean | Std. Dev. | Median | Min. | Max. | Mean | Std. Dev. | Median | Min. | Max. |
| Over-pledging ($) | 2,415.78 | 12,580.85 | 755 | 0 | 843,433 | 515.90 | 1,729.14 | 112 | 0 | 45,022 |
| Over-pledging/Total pledge amount | 0.19 | 0.20 | 0.12 | 0 | 1 | 0.19 | 0.22 | 0.10 | 0 | 1 |
| Number of OP backers | 35.91 | 118.50 | 16 | 0 | 4,047 | 6.95 | 12.84 | 4 | 0 | 392 |
| Proportion of OP backers | 0.21 | 0.17 | 0.15 | 0 | 1 | 0.24 | 0.20 | 0.19 | 0 | 1 |

*Note*: The number of successful projects and failed projects are equal to 6,962 and 4,257, respectively.

**Figure 3 Differences in Over-pledging Spectrum (Successful vs. Failed Projects)**




(a) PDF (b) CDF

The over-pledging spectra for successful and failed projects are plotted in Figures 3(a) and (b) for PDF and CDF, respectively. The shapes of the distributions are similar, but the OP spectrum of successful projects shifts rightward compared to that of failed projects. There exists a higher proportion of failed projects having an OP score greater than 0.6, resulting in the average of OP scores not significantly different between successful and failed projects, as shown in Table 4.

*Over-pledging spectrum by project state.* Table 5 summarizes the over-pledging amount and the

number of over-pledged backers in the periods before and after reaching the funding target.[4] Over-pledging decreased significantly after reaching the funding target, in both the volume and intensity. After reaching the funding target, the average amount of over-pledging decreased from $1,426.08 to $988.60, and the average over-pledging as a proportion of total pledge amount decreased significantly from 0.19 to 0.12. Similarly, the average proportion of backers who are willing to over-pledge decreased from 0.20 to 0.17 after the funding target was reached. Existing studies have documented that backers of crowdfunding projects have a stronger motivation to pledge before reaching the funding target, especially when facing an all-or-nothing mechanism (Kuppuswamy and Bayus 2017, Dai and Zhang 2019). Our results show that the intensity of over-pledging is also significantly higher before reaching the target. Having said that, over-pledging behavior continued after reaching the targets and still constituted a significant part of the total pledge amount. The above result suggests that the strategic motivation of reaching the targets can be one of the reasons motivating the over-pledging behavior, but it is certainly not the only one.

**Table 5 Comparison of Over-pledging Before and After Reaching the Funding Target**

| | Before | | | | | After | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | Mean | Std. Dev. | Median | Min. | Max. | Mean | Std. Dev. | Median | Min. | Max. |
| Over-pledging ($) | 1,426.08 | 2,702.9 | 500 | 0 | 35,800 | 988.60 | 12,550.75 | 54.87 | 0 | 842,498 |
| Over-pledging/Total pledge amount | 0.19 | 0.20 | 0.11 | 0 | 1 | 0.12 | 0.21 | 0.03 | 0 | 1 |
| Number of OP backers | 17.81 | 23.81 | 11 | 0 | 597 | 18.83 | 115.18 | 3 | 0 | 4,044 |
| Proportion of OP backers | 0.20 | 0.17 | 0.15 | 0 | 1 | 0.17 | 0.22 | 0.09 | 0 | 1 |

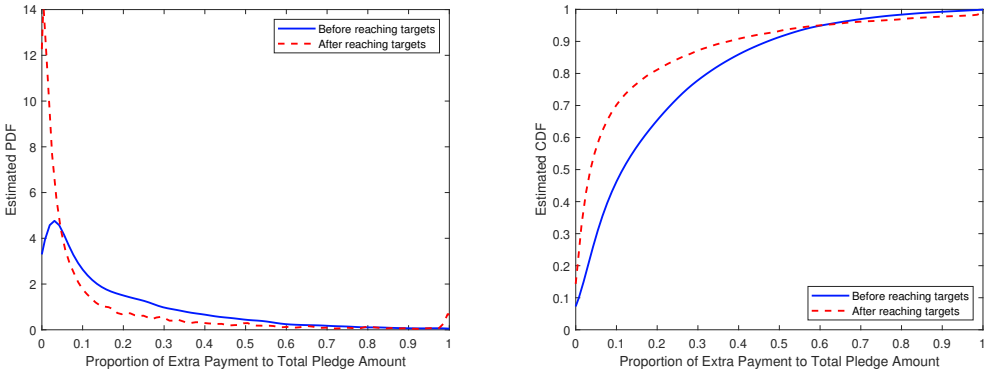
*Note*: Here the total amount is calculated within each period, either before or after reaching the funding target.

Figure 4 shows the OP spectra before and after reaching the funding targets. In particular, the OP spectrum before (after) reaching the funding target is given by the ratio between the over-pledging and the total pledge amount in respective periods. The plots suggest that the OP spectrum before reaching the funding target tends to have a first-order stochastic dominance over that after reaching the funding target. That is, the intensity of over-pledging before reaching the funding target is consistently higher than that after.

*Over-pledging spectrum by product category.* Kickstarter classified all projects into 15 product categories. The summary statistics of the over-pledging spectrum of the projects in each category are summarized in Table 6. The categories are presented in the descending order of the mean score

[4] Note that only the successful projects are considered in this *before-after* comparison. In addition, 484 successful projects that did not receive any pledges after reaching the funding targets are excluded from the analysis, as their intensity of over-pledging cannot be computed.

**Figure 4 Differences in Over-pledging Spectrum (Before vs. After Reaching Targets)**




(a) PDF (b) CDF

of the corresponding OP spectrum. In addition, the OP spectrum for each category is summarized using boxplots in Figure 5.

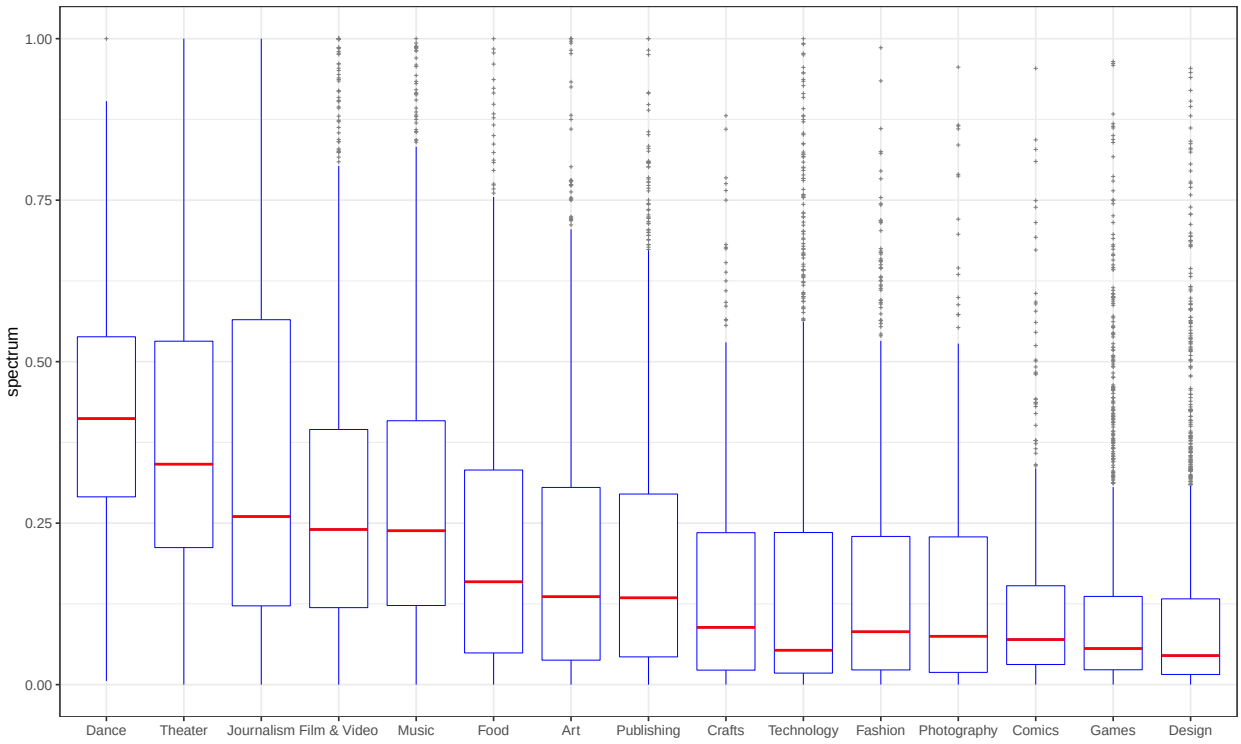
**Table 6 Over-pledging Spectrum by Project Category**

| | Count | Mean | Std. Dev. | Median | Minimum | Maximum |
|---|---|---|---|---|---|---|
| Dance | 82 | 0.43 | 0.21 | 0.41 | 0.01 | 1 |
| Theater | 221 | 0.37 | 0.22 | 0.34 | 0 | 1 |
| Journalism | 93 | 0.35 | 0.28 | 0.26 | 0 | 1 |
| Film & Video | 1,219 | 0.29 | 0.22 | 0.24 | 0 | 1 |
| Music | 1,137 | 0.29 | 0.22 | 0.24 | 0 | 1 |
| Food | 599 | 0.22 | 0.22 | 0.16 | 0 | 1 |
| Art | 650 | 0.21 | 0.22 | 0.14 | 0 | 1 |
| Publishing | 1,163 | 0.20 | 0.21 | 0.13 | 0 | 1 |
| Crafts | 266 | 0.17 | 0.20 | 0.09 | 0 | 0.88 |
| Technology | 1,152 | 0.16 | 0.22 | 0.05 | 0 | 1 |
| Fashion | 703 | 0.16 | 0.19 | 0.08 | 0 | 0.99 |
| Photography | 247 | 0.16 | 0.20 | 0.07 | 0 | 0.96 |
| Comics | 522 | 0.12 | 0.14 | 0.07 | 0 | 0.95 |
| Games | 1,680 | 0.11 | 0.15 | 0.06 | 0 | 0.96 |
| Design | 1,485 | 0.11 | 0.16 | 0.04 | 0 | 0.95 |

*Note:* The total number of projects is 11,219.

Both Table 6 and Figure 5 show that the category-level OP spectrum varies considerably across product categories. Specifically, on the top, the *dance* category has the highest average score with around 43% of the total pledge amount as over-pledging. In addition, four other categories, including *theater*, *journalism*, *film & video* and *music*, all have an average score above 0.25. At the other end, the average OP score is the lowest in the categories such as *design*, *games*, and *comics*.

**Figure 5 Boxplot of Over-pledging Spectrum by Project Category**




*Note.* The red line indicates the median of the OP scores, the lower bound of the box indicates the first quartile ($Q_1$/25th percentile), and the upper bound of the box indicates the third quartile ($Q_3$/75th percentile). A project with OP score $< Q_1 - 1.5IQR$ or $> Q_3 + 1.5IQR$, where $IQR = Q_3 - Q_1$, is marked as a cross in the plot.

In these categories the average OP score was below 0.15.

The difference in the OP spectrum across product categories is consistent with product characteristics. Overall, the crowdfunding projects in the fine arts categories rank high in the over-pledging intensity. These products, e.g., a show or a film, are often in the nature of public goods and thus more likely to induce altruism from backers. Albeit in a different context, this phenomenon is likewise evident in Gneezy et al. (2010), which suggests an increased propensity among customers to make greater contributions towards philantrhopic endeavors. Conversely, a typical project in *games* or *design* usually results in a product intended for individual consumption, making it less likely to foster a sense of altruism.

*Profiles of over-pledging clusters.* We rank all projects based on the OP score and divide them into 3 clusters of equal numbers: the top, middle, and bottom third. We then create profiles to describe each cluster based on OP scores, project and creator characteristics, campaign dynamics, and project outcomes. We summarize the distribution of OP scores within each cluster in Table 7. The average OP scores for the top one-third of projects was 43%. The top cluster also shows the

greatest variation, where OP scores ranged from 0.21 to 1. On the other end of the spectrum, the average OP score in the bottom one-third of projects was only 2% and ranged from 0 to 5%.

**Table 7 Over-Pledging Spectrum at the Cluster Level**

| | Count | Mean | Std. Dev. | Median | Minimum | Maximum |
|---|---|---|---|---|---|---|
| Top 1/3 | 3,740 | 0.43 | 0.19 | 0.38 | 0.21 | 1 |
| Middle 1/3 | 3,739 | 0.12 | 0.05 | 0.11 | 0.05 | 0.21 |
| Bottom 1/3 | 3,740 | 0.02 | 0.02 | 0.02 | 0 | 0.05 |

*Note*: The total number of projects is 11,219.

**Table 8 Cluster Mean of Each Profile Variable**

| | Top 1/3 | Middle 1/3 | Bottom 1/3 |
|---|---|---|---|
| **Project Characteristics** | | | |
| Funding target ($) | 13,435.24 | 16,155.71 | 17889.31 |
| Duration (hours) | 774.36 | 760.17 | 777.79 |
| Number of reward levels | 8.52 | 10.65 | 9.87 |
| Video (= 1 if yes; = 0 otherwise) | 0.71 | 0.73 | 0.70 |
| **Creator Characteristics** | | | |
| First time (= 1 if yes; = 0 otherwise) | 0.83 | 0.72 | 0.63 |
| Number of projects backed before | 2.83 | 8.44 | 8.98 |
| Number of collaborators | 0.43 | 0.49 | 0.42 |
| US (= 1 if yes; = 0 otherwise) | 0.71 | 0.67 | 0.61 |
| **Campaign Dynamics** | | | |
| Update (# per hour) | 0.01 | 0.01 | 0.01 |
| Share (# per hour) | 0.01 | 0.03 | 0.03 |
| **Outcome** | | | |
| Total number of backers | 66.7 | 208.2 | 464.13 |
| Total pledge amount ($) | 5,748.35 | 19,548.22 | 35,941.26 |
| Number of OP backers | 20.42 | 30.73 | 23.46 |
| Over-pledging ($) | 2,282.84 | 1,955.95 | 845.92 |
| Over-pledging/Total pledge amount | 0.43 | 0.12 | 0.02 |

*Note*: This table summarizes the mean values in Tables A.2 to A.4 of the Appendix.

We summarize the mean of each profile variable at the cluster level in Table 8. In addition to variables used for sample summary statistics in Table 1, we include two more variables on campaign dynamics. These variables measure creator/backer engagement throughout crowdfunding campaigns. Specifically, "update" indicates the frequency a project creator posted project updates, measured by the average number of updates per hour. These updates range from simple reminders/expressions of appreciation (e.g., a thank-you note) to amendments to the product

or service they seek to deliver (e.g., a new design or feature for a product). The second variable "share" measures the average number of times per hour the link of a project was shared on social networks, including Facebook and Twitter. (Detailed summary statistics are relegated to the e-companion.) Based on the statistics summarized in Table 8, we present project profiles for each cluster as follows.

*(a) Project profile of the top cluster.* A typical project in the top cluster has a smaller target and fewer reward levels than the other two clusters. The creator of the project is more likely to come from the U.S., and more likely to be a novice with less experience on the platform in terms of creating and backing projects. The project was shared less often on social networks.

*(b) Project profile of the middle cluster.* A typical project in this cluster has a medium-size target but the highest number of reward levels. The project creator may not have experience launching crowdfunding campaigns on the platform but has backed a few projects before. The project attracted moderate attention on social networks.

*(c) Project profile of the bottom cluster.* A typical project in this cluster tends to be relatively expansive. The project's creator usually boasts considerable experience in either creating or backing prior projects. Furthermore, it's less common for the creator to be based in the U.S.

## 3. Over-Pledging: With vs. Without Reward

As mentioned before, a backer who over-pledges may request a reward or choose to pledge without a reward; for ease of exposition, we refer to the former type of such backers as *funders* and the latter type as *donors*. Up to this point, our analysis has treated over-pledging as a unified category, encompassing both funders and donors. We will now assess these two types of over-pledged backers individually, recognizing that their motivations for voluntary contributions may differ significantly. In crowdfunding, donors are not fundamentally purchasing products; instead, they extend financial contributions to endorse the innovators behind these projects. In several regards, these donors align closely with contributors making philanthropic donations, whose motivations often encompass various forms of altruism (both pure and impure) along with other strategic considerations, as delineated in the literature. It is plausible that these factors also influence the over-pledging tendency among funders. However, there is one salient difference between donors and funders, wherein a product or service exchange occurs only for funders, but not for donors. An abundance of empirical evidence in the context of Pay-What-You-Want (PWYW) suggests that customers are inclined to contribute more voluntarily as their perceived value of the product/service increases, facilitating a more equitable distribution of this surplus between customers and sellers. This particular

concept of fairness is frequently encapsulated in the term "inequity aversion" as posited by Fehr and Schmidt (1999). By juxtaposing the behavioral patterns of funders with those of donors, we can discern whether inequity aversion significantly influences funders' propensity for over-pledging, and how their willingness-to-pay above the required price for the reward, which is referred to as *perceived value surplus* in the rest of this paper, is affected by the characteristics of creators and their respective projects.

**Table 9 Over-pledging from Funders and Donors**

| | Funders | | | | | Donors | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | Mean | Std. Dev. | Median | Min. | Max. | Mean | Std. Dev. | Median | Min. | Max. |
| Payment ($, by funders/donors) | 2,139.05 | 20,699.27 | 320 | 0 | 1,779,178 | 800.38 | 2,408.87 | 136.07 | 0 | 95,091 |
| Payment/Total pledge amount | 0.14 | 0.16 | 0.09 | 0 | 1 | 0.12 | 0.17 | 0.04 | 0 | 1 |
| Over-pledging ($) | 894.50 | 8,663.65 | 120 | 0 | 748,342 | 800.38 | 2,408.87 | 136.07 | 0 | 95,091 |
| Over-pledging/Total pledge amount | 0.07 | 0.11 | 0.03 | 0 | 1 | 0.12 | 0.17 | 0.04 | 0 | 1 |
| Number of funders or donors | 15.70 | 78.91 | 4 | 0 | 3,462 | 9.17 | 25.05 | 4 | 0 | 1,118 |
| Proportion of funders or donors | 0.10 | 0.10 | 0.07 | 0 | 1 | 0.12 | 0.14 | 0.07 | 0 | 1 |

*Note*: *Payment* refers to the total amount pledged by funders (donors) per project. The other variables are defined as the same as in Table 2.

Table 9 summarizes the number and payment amount of funders and donors. Here *payment* refers to the total amount pledged by funders (donors) in a project. The table shows that, first, over-pledging behavior is a significant part of payments from the funders. For the average payment of $2,139.05 from the funders for each project, $894.5 (42%) is the extra payment, and the rest 58% represents the required payment for the reward. Second, on average, there were more funders than donors (15.7 versus 9.17 per project) and more over-pledged contributions from the funders than from the donors ($894.5 versus $800.38 per project), providing preliminary evidence for a positive effect of fairness concern on the over-pledging behavior. Overall, the funders have a stronger direct impact on project success than the donors.

Next, we compare the over-pledging behavior between successful and failed projects, for funders and donors separately. According to Table 10, successful projects received a higher amount of over-pledging from a larger number of backers, including both funders and donors, than failed projects. This observation is consistent with the results in Table 4. Yet, a more intriguing finding emerges when examining the difference in average over-pledging amounts between funders and donors in relation to the project's success or failure. In successful projects, funders' average over-pledging amount stands at approximately $1317.62, a figure that precipitously drops to roughly $202.52

**Table 10 Comparing Extra Payment Between Successful and Failed Projects by Funders and Donors**

| | Successful Projects | | | Failed Projects | | | Two-sample t-test | |
|---|---|---|---|---|---|---|---|---|
| | Mean | Std. Dev. | Median | Mean | Std. Dev. | Median | t-stat. | p-value |
| **Funders** | | | | | | | | |
| Payment ($) | 3212.3 | 26198.98 | 660.86 | 383.83 | 1317.84 | 70 | 8.99 | 0.000 |
| Payment/Total pledge amount | 0.15 | 0.15 | 0.1 | 0.13 | 0.18 | 0.06 | 7.25 | 0.000 |
| Over-pledging ($) | 1317.62 | 10953.85 | 255 | 202.52 | 906.4 | 21.36 | 8.45 | 0.000 |
| Over-pledging/Total pledge amount | 0.07 | 0.11 | 0.04 | 0.07 | 0.12 | 0.02 | 3.29 | 0.000 |
| Number of funders | 23.45 | 99.26 | 8 | 3.02 | 6.43 | 1 | 17.11 | 0.000 |
| Proportion of funders | 0.1 | 0.09 | 0.08 | 0.09 | 0.1 | 0.06 | 6.89 | 0.000 |
| **Donors** | | | | | | | | |
| Over-pledging ($) | 1098.16 | 2856.38 | 267.79 | 313.38 | 1252.2 | 40.31 | 20.00 | 0.000 |
| Over-pledging/Total pledge amount | 0.12 | 0.17 | 0.04 | 0.12 | 0.18 | 0.03 | -1.29 | 0.197 |
| Number of donors | 12.38 | 30.67 | 6 | 3.93 | 8.4 | 2 | 21.70 | 0.000 |
| Proportion of donors | 0.1 | 0.13 | 0.05 | 0.15 | 0.16 | 0.1 | -15.89 | 0.000 |
| Proportion of top donors to donors | 0.8 | 0.29 | 1 | 0.7 | 0.39 | 1 | 12.94 | 0.000 |

*Note*: (1) The numbers of successful and failed projects are equal to 6,962 and 4,257, respectively. (2) The last two columns are results of t-tests for the difference of means between successful and failed projects. (3) Top donors are those whose pledges were high enough to qualify a reward, but they donated without requiring any reward.

in failed projects - a differential nearing $1115.1. In contrast, the average over-pledging amount from donors diminishes by approximately $784.78. The significantly larger decrease in funders' over-pledging contributions between successful and failed projects provides another set of fresh evidence on the potential impact of fairness on funders' pledging propensity. Such a discrepancy between funders and donors may occur if failed projects, on average, deliver lower quality products or services, or if the failure of a project serves as an indicator of inferior product or service quality, thereby leading to a diminished perceived value.

Table 11 compares the over-pledging before and after reaching the funding target for the funders and donors separately. Consistent with aggregate results summarized in Table 5, the intensity of over-pledging, measured by extra payment as a fraction of the total pledge amount, was higher before reaching funding targets than after, for both funders and donors. However, funders and donors demonstrated markedly divergent reactions, in terms of amount of over-pledging, to the achievement of funding targets. On average, donors contributed significantly more prior to the attainment of these goals, with an approximate difference of $620.62 in average over-pledging amounts before and after reaching the thresholds. As a project successfully meets its funding goal, the impetuses for donors — such as altruism and/or strategic considerations to facilitate project success — may diminish in relevance, or vanish altogether. This dynamic leads to considerably reduced over-pledging amounts from donors post-threshold. Conversely, funders' contributions

**Table 11 Comparing Over-pledging Before and After Reaching the Funding Target by Funders and Donors**

| | Before | | | After | | | Two-sample t-test | |
|---|---|---|---|---|---|---|---|---|
| | Mean | Std. Dev. | Median | Mean | Std. Dev. | Median | t-stat. | p-value |
| **Funders** | | | | | | | | |
| Payment ($) | 1,231.62 | 2,438.59 | 456.92 | 2,059.26 | 26,626.40 | 52 | -2.49 | 0.013 |
| Payment/Total pledge amount | 0.15 | 0.16 | 0.10 | 0.10 | 0.17 | 0.04 | 17.47 | 0.000 |
| Over-pledging ($) | 578.61 | 1,385.05 | 172 | 761.74 | 11,122.74 | 15 | -1.31 | 0.189 |
| Over-pledging/Total pledge amount | 0.07 | 0.11 | 0.04 | 0.04 | 0.10 | 0.01 | 16.81 | 0.000 |
| Number of funders | 9.81 | 14.56 | 6 | 14.49 | 98.77 | 1 | -3.77 | 0.000 |
| Proportion of funders | 0.10 | 0.10 | 0.08 | 0.08 | 0.13 | 0.04 | 11.62 | 0.000 |
| **Donors** | | | | | | | | |
| Over-pledging ($) | 847.48 | 2,028.68 | 165.42 | 226.86 | 1,781.74 | 10 | 18.49 | 0.000 |
| Over-pledging/Total pledge amount | 0.11 | 0.16 | 0.03 | 0.08 | 0.19 | 0 | 10.36 | 0.000 |
| Number of donors | 8 | 13.30 | 4 | 4.34 | 23.64 | 1 | 10.85 | 0.000 |
| Proportion of donors | 0.10 | 0.12 | 0.05 | 0.09 | 0.18 | 0.01 | 3.28 | 0.001 |
| Proportion of top donors to donors | 0.82 | 0.28 | 1 | 0.72 | 0.38 | 1 | 14.06 | 0.000 |

*Note*: 6,468 out of 6,710 successful projects are considered in this table, as the remaining 487 successful projects did not receive any pledge after reaching their funding target. In addition, the total pledge amount is calculated based on the corresponding period, i.e., either before or after the funding target was reached. The same applies to the calculation of the proportion of funders (donors).

demonstrated an increase after the attainment of the target, with an average difference in over-pledging before and after thresholds of -$183.13 (the difference is not statistically significant). This contrast in behavior between funders and donors can also be attributable to changes in funders' perceived fair value of the product or service on offer. Successful fundraising to meet a project's target can serve as an indicator of the quality of the product or service, which results in an enhanced perceived fair value, offsetting the diminished influence of other motivating factors, such as altruism and strategic considerations.

In summary of the findings, over-pledging contributions were sizable and exerted a significant influence on the success of crowdfunding projects. Notably, two distinct categories of backers who over-pledge have been identified: funders who over-pledge in return for a reward, and donors who over-pledge without expectation of a reward. In alignment with extant literature, the motives of donors could encompass both pure and impure forms of altruism, as well as strategic considerations. This supposition is substantiated by notable contributions from donors observed in unsuccessful projects, and in the phase prior to reaching the funding target among successful projects. Conversely, for funders, while altruism and strategic considerations also apply, fairness also emerges as a significant determinant in their over-pledging behavior. This is evidenced by a markedly divergent treatment effect observed in successful projects and/or those having achieved their funding

targets, relative to the donors. In the subsequent section, we aim to undertake a more comprehensive analysis to elucidate how the motivations of funders and donors are affected by additional correlates linked to the project creator and the project itself. This further study will enable a deeper understanding of the complex dynamics of over-pledging within crowdfunding platforms.

## 4. Correlates of Over-Pledging

In this section, we investigate correlates of the over-pledging scores (OP scores). Our examination delves into an array of associated factors, embracing both static and dynamic characteristics, as well as variables that originate from within the platform alongside those off-platform factors. The objectives of this analysis are bifurcated: (1) Prior literature alludes to various potential motivations for voluntary contributions from donors, ranging from (pure or impure) altruism to more strategic considerations. Our analysis will elucidate how these motivating factors could be swayed by correlates at the creator and project levels; (2) We disentangle fairness from other motivations through a difference-in-difference analysis between funders and donors, which sheds light on how it may be influenced by the characteristics inherent to the creator and the project.

*Correlations Between OP Score and Project/Creator Characteristics*

We first examine the correlation between a project's OP score and a number of project- and creator-related factors. We run OLS regressions to quantify the magnitude of those relationships and the differences between funders and donors. For dependent variables, in addition to the overall project-level OP score, we consider the OP score by funders, by donors, and their score difference, separately. The OP score by funders (donors) is the total over-pledging from funders (donors) as a fraction of the total pledge amount of the project.

We estimate the following regression specification:

$$\begin{aligned} OP\ score_i = {} & \beta_0 + \beta_1 log(target_i) + \beta_2 reward_i + \beta_3 video_i + \beta_4 first\ time_i + \beta_5 backed_i \\ & + \beta_6 collab_i + \beta_7 US_i + \beta_8 update_i + \beta_9 share_i + category_i + launch\ time_i + \epsilon_i, \quad (1) \end{aligned}$$

where $log(target_i)$ is the natural log of the funding target of project $i$. Taking the log narrows the range of the variable, and alleviates the skewness in the distribution of funding targets. $reward_i$ represents the number of reward levels for project $i$. $video_i$ is a dummy variable which equals 1 if at least one video was posted on the web page of the crowdfunding campaign and 0 otherwise. These three variables describe the project characteristics. For the demographic information of project creators, $first\ time_i$ equals 1 if the creator of project $i$ had never launched any project on Kickstarter before, and 0 otherwise. $backed_i$ represents the number of projects a project creator

had backed before. $collab_i$ refers to the number of people in the project creation team, excluding the lead creator, for project $i$. In the sample, the number of collaborators varied from 0 to 13, with 0 implying that the crowdfunding campaign was created by one single creator. $US_i$ is equal to 1 if the project creator resided in the U.S., and 0 otherwise. We also include two variables on backer engagement in the model. $update_i$ measures how frequently project $i$ was updated throughout the crowdfunding campaign, and $share_i$ measures the frequency project $i$ was shared on social network platforms. Lastly, we control for category fixed effects $category_i$ and launch time fixed effects $launch\ time_i$, the latter referring to the month when the project was launched.

The regression results are summarized in Tables 12 and 13. In particular, columns (1), (2) and (3) show the results of the overall OP score, the OP scores by funders and by donors as dependent variables, respectively, and column (4) shows the results with the difference of OP scores between funders and donors as the dependent variable. Table 12 summarizes the effects of explanatory variables, whereas Table 13 shows the category fixed effects.

We will mainly focus on results in columns (3) and (4) of the tables, which illustrate how the motives of donors, such as altruism and strategic considerations, and the perceived fair values are correlated with creator/project characteristics, respectively. The OP score is positively correlated with the project size for donors ($\beta_{log(target)} \approx 0.002,\ p - value < 0.05$), meaning that donors on average will contribute more as the funding target increases. A larger project presents a greater need from the creater, which may provoke greater empathy from donors, which has been shown as an important source of alturistic motivation in the psychology literature (Batson et al. 1981, Cialdini et al. 1987). On the contrary, compared to donors, the funders' over-pledging intensity decreases in bigger projects ($\beta_{log(target)} \approx -0.006, p - value < 0.01$), indicating smaller value surplus from funders as the project size increases.

A higher number of *reward levels* is related to a lower OP score from donors. Given all others the same, a higher number of reward levels implies smaller gaps between the prices of adjacent reward levels. This disincentise donors to over-pledge as the lower reward levels present themselves as attractive alternatives. On the contrary, the number of reward levels does not affect the perceived value surplus of the products/service significantly.

The OP score of projects by *first-time creators* are significantly higher than those created by experienced creators for donors. As previously noted in our analysis, 73% of the projects in our sample were initiated by first-time creators, individuals who had no prior experience in launching crowdfunding campaigns on Kickstarter. This lack of experience among first-time creators is a mixed blessing. It can often translate into a greater need for funding, potentially eliciting a higher

**Table 12 Regression Results on Over-Pledging Scores (i): Explanatory Variables**

| | (1) | (2) | (3) | (4) |
|---|---|---|---|---|
| | OP Score | OP Score (funders) | OP Score (donors) | OP Score (funders-donors) |
| log(target) | −0.001 | −0.003*** | 0.002** | −0.006*** |
| | (0.001) | (0.001) | (0.001) | (0.001) |
| reward (# of reward levels) | −0.005*** | −0.003*** | −0.002*** | −0.000 |
| | (0.001) | (0.000) | (0.000) | (0.000) |
| video (dummy) | −0.008* | −0.011*** | 0.003 | −0.014*** |
| | (0.004) | (0.003) | (0.004) | (0.005) |
| first time (dummy) | 0.049*** | 0.015*** | 0.034*** | −0.019*** |
| | (0.004) | (0.002) | (0.003) | (0.004) |
| backed (# of projects backed) | −0.000*** | −0.000 | −0.000*** | 0.000*** |
| | (0.000) | (0.000) | (0.000) | (0.000) |
| collab (# of collaborators) | −0.000 | −0.001 | 0.001 | −0.002 |
| | (0.002) | (0.001) | (0.002) | (0.002) |
| US (dummy) | 0.026*** | 0.002 | 0.025*** | −0.023*** |
| | (0.004) | (0.002) | (0.003) | (0.004) |
| update (# of updates per hour) | −0.935*** | 0.062 | −0.997*** | 1.060*** |
| | (0.127) | (0.070) | (0.089) | (0.097) |
| share (# of shares per hour) | −0.037** | −0.012*** | −0.025** | 0.013 |
| | (0.016) | (0.005) | (0.012) | (0.008) |
| Category fixed effects | Yes | Yes | Yes | Yes |
| Launch month fixed effects | Yes | Yes | Yes | Yes |
| Observations | 11,219 | 11,219 | 11,219 | 11,219 |
| Adjusted $R^2$ | 0.173 | 0.043 | 0.169 | 0.090 |

*Note:* Category fixed effects are detailed in Table 13. Standard errors clustered by project are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

degree of sympathy from potential donors. However, the lack of experience often results in suboptimal project design, thereby leading to a diminished perceived value surplus. This is evidenced by the markedly reduced intensity of over-pledging from funders.

Projects by *U.S.-based creators* received a higher proportion of over-pledging from donors, likely due to geographic proximity. We do not have demographic information about the backers in our

**Table 13 Regression Results for Over-Pledging Scores (ii): Category Fixed Effects**

| | (1) OP Score | (2) OP Score (funders) | (3) OP Score (donors) | (4) OP Score (funders-donors) |
|---|---|---|---|---|
| Dance | 0.290*** | 0.030** | 0.260*** | −0.230*** |
| | (0.023) | (0.014) | (0.021) | (0.028) |
| Theater | 0.247*** | 0.014* | 0.233*** | −0.220*** |
| | (0.015) | (0.008) | (0.015) | (0.018) |
| Journalism | 0.210*** | 0.035** | 0.174*** | −0.139*** |
| | (0.028) | (0.015) | (0.026) | (0.031) |
| Film & Video | 0.165*** | 0.023*** | 0.142*** | −0.119*** |
| | (0.007) | (0.004) | (0.006) | (0.008) |
| Music | 0.165*** | 0.035*** | 0.130*** | −0.096*** |
| | (0.008) | (0.004) | (0.006) | (0.008) |
| Food | 0.096*** | 0.001 | 0.095*** | −0.094*** |
| | (0.010) | (0.005) | (0.008) | (0.010) |
| Art | 0.090*** | 0.006 | 0.083*** | −0.077*** |
| | (0.010) | (0.005) | (0.008) | (0.010) |
| Publishing | 0.083*** | 0.025*** | 0.059*** | −0.034*** |
| | (0.007) | (0.004) | (0.006) | (0.007) |
| Crafts | 0.051*** | 0.014 | 0.037*** | −0.024* |
| | (0.013) | (0.009) | (0.010) | (0.013) |
| Technology | 0.042*** | 0.012*** | 0.030*** | −0.018** |
| | (0.008) | (0.004) | (0.006) | (0.007) |
| Fashion | 0.040*** | 0.003 | 0.037*** | −0.034*** |
| | (0.008) | (0.005) | (0.007) | (0.008) |
| Photography | 0.042*** | −0.008 | 0.049*** | −0.057*** |
| | (0.013) | (0.007) | (0.011) | (0.013) |
| Comics | 0.040*** | 0.014*** | 0.025*** | −0.011 |
| | (0.008) | (0.005) | (0.006) | (0.008) |
| Games | 0.016*** | 0.004 | 0.012*** | −0.008 |
| | (0.006) | (0.004) | (0.005) | (0.006) |

*Note:* This table complements Table 12 by illustrating category fixed effects in the regressions. *Design* is the base category. Standard errors clustered by project are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

sample. However, based on internet traffic data, around 35% of visitors came from U.S.,[5] which is much higher than any other regions in the world. The close geographical proximity or cultural similarity between the backers and project creators may foster the stronger over-pledging intensity from donors for the U.S.-based projects. In a different setting of online lending, Burtch et al. (2014) and Lin and Viswanathan (2016) observe a similar phenomenon that lenders prefer culturally sim-

[5] Source: https://www.alexa.com/siteinfo/kickstarter.com; https://www.similarweb.com/website/kickstarter.com.

ilar and geographically proximate borrowers. However, the perceived value surplus is significantly lower from projects by U.S.-based creators.

Interestingly, a negative relationship is observed between the frequecy of *project updates* and the project's OP score by donors. Frequent updates may potentially signify the capability of project creators. Moreover, projects with frequent updates are more likely to have achieved success, which, in turn, might attenuate the altruistic inclinations of the donors due to the reduced perception of need. On the other hand, the inherent value of the product or service tends to augment following several rounds of updates, resulting in an elevation of the perceived value surplus.

The number of *social network shares* is negatively related to a project's OP score for donors. This is surprising because online social media is believed to be effective in coordinating consumer payments to sustain pay-what-you-want campaigns (e.g., Mak et al. 2015). Whereas, our result implies that more shares on social media may backfire for backers' intention to over-pledge. However, this observation may suffer by endogeneity issues, as creators who share more frequently on social networks might already be experiencing a lack of pledging or over-pledging from the outset. In a subsequent analysis, we will leverage an exogenous shock on Kickstarter to demonstrate that increased social network activities can indeed lead to a higher intensity of over-pledging from donors, but not from funders (see Table 15).

Table 13 shows the category fixed effects for OP scores, column (1) for all projects, columns (2) and (3) for funders and donors separately, and column (4) for their score difference. Column (1) shows that, first, the overall product category effects are consistent with earlier discussions. Specifically, there is a substantial difference across the categories, from 1.6% for the *games* category to 29.0% for the *dance* category, indicating a large range of 27.4%. Second, all the categories related to fine arts, including *dance*, *theater*, *journalism*, *film & video*, and *music*, enjoyed very high category-specific effects of OP scores. On the other hand, those categories related to commercial products, like *technology*, *fashion*, *comics*, and *games*, had a very small amount of over-pledging. The results for funders and donors (columns (2) and (3), respectively) indicate the importance of donors in driving the variations across product categories. In column (2), for the over-pledging from the funders, although variations existed across some categories, the fixed-effect estimates are small in magnitude, ranging from zero to 3.5%. On the other hand, in column (3), for the over-pledging from the donors, the variations across categories are substantial. Comparing columns (1) and (3), approximately 90% of all the category-specific variations in OP scores can be attributed to the over-pledging from the donors.

As previously deliberated, the motivations of donors may incorporate varying degrees of altruism (either pure or impure) as well as additional strategic considerations. These could include aiding in the success of projects or engaging in reciprocal acts. Many of these motives are influenced by the perceived need of the projects. A higher level of need, as indicated by a significant gap between the pledged amount and the funding goal, or the creator's lack of experience, may stimulate empathetic or sympathetic responses from donors, thereby resulting in an intensified level of over-pledging. This proposition aligns with the findings evident in our regression analysis. The data suggests that the intensity of over-pledging from donors is typically higher for larger-scale projects, projects undertaken by novice creators, and projects managed by creators based in the United States. The aforementioned motivational influences also permeate the decision-making processes of funders. However, the dynamics differ due to the transactional exchange that exists between funders and project creators. Funders may pledge additional financial support driven by their positive perceived surplus value, while simultaneously aspiring towards an equitable allocation of this surplus between themselves and project creators. Our empirical analysis strongly corroborates the proposition that inequity aversion plays a significant role in shaping the over-pledging intensity among funders, as demonstrated by the distinctive impacts of creator and project characteristics on the over-pledging behavior of donors versus funders. On the whole, the perceived value surplus tends to be higher for smaller-scale projects, projects that do not include video content, projects managed by experienced and/or non-U.S. based creators, and projects characterized by frequent updates. A heightened perceived value surplus may stem from a lack of proficiency on the part of project creators in accurately setting prices for reward tiers, which we posit could be particularly applicable to smaller projects and those lacking video content. Conversely, it might also be a result of augmented value creation by creators, likely in projects helmed by experienced creators and/or projects with frequent updates. As we will elucidate later in this paper, under certain conditions, we are able to disentangle these causal factors and precisely identify the basis for the increased perceived value surplus. However, for a substantial proportion of scenarios, our dataset does not permit a clear segregation of these causes. These intricate explorations warrant further scrutiny and are left for future research undertakings.

### 4.1. Impact of Funding Progress on OP Score

Findings gleaned from Table 11 offer some initial indications that the responses of funders and donors to the attainment of funding targets diverge significantly. Specifically, the amounts over-pledged by donors diminish notably once the threshold is crossed, while those pledged by funders exhibit an increase. Nonetheless, as this variance is computed at an aggregate level, we cannot

categorically exclude the presence of numerous confounding factors. In this section, we embark on a more rigorous difference-in-difference (DiD) regression analysis to delve into this effect more systematically.

We construct a panel data set at the daily level. The key variables of interest are the OP scores either by funders or donors, i.e., $OP\ score_{it}$, which are calculated as the ratio of over-pledging from funders and donors to the total daily pledged amount for project $i$ on day $t$, respectively. We run the following regression to estimate the difference of OP scores between funders and donors after (vs. before) the funding target:

$$OP\ score_{it} = \beta_0 + \beta_1 Funder + \beta_2 AfterSuccess_{it} + \beta_3 Funder \times AfterSuccess_{it} + FEs + \epsilon_{it}. \quad (2)$$

$Funder$ is a dummy variable indicating whether or not an observation is from a funder ($=1$ for a funder and $=0$ for a donor). $AfterSuccess_{it}$ is another dummy variable, which $=1$ if project $i$ has reached its funding target on day $t$, and $=0$ otherwise. $FEs$ account for project and time (i.e., year and month) fixed effects. Note that this analysis only includes successful projects as there is no post-threshold period for failed projects.

**Table 14 Results of DiD Regression.**

| | OP Score |
|---|---|
| $Funder$ | $-0.004^{***}$ |
| | (0.000) |
| $AfterSuccess$ | $-0.000$ |
| | (0.000) |
| $Funder \times AfterSuccess$ | $0.003^{***}$ |
| | (0.000) |
| Project fixed effects | Yes |
| Time fixed effects | Yes |
| Observations | 327,518 |
| Adjusted $R^2$ | 0.061 |

*Note*: Standard errors clustered by project are reported in parentheses. $^{***}$ $p<0.01$, $^{**}$ $p<0.05$, $^{*}$ $p<0.1$.

The results are summarized in Table 14. The key variable of interest is the interaction term $Funder \times AfterSuccess_{it}$, whose coefficient (i.e., $\beta_3$) represents the change in OP score by funders relative to that by donors after (vs. before) the funding target was reached. Consistent with preliminar results presented in Table 11, the coefficient of the interaction term $Funder \times AfterSuccess_{it}$ (i.e., $\beta_3 \approx 0.003$, $p-vale < 0.01$) is positive and statistically significant, suggesting that the shift

in over-pledging intensity from funders after crossing the threshold (as compared to pre-threshold levels) is significantly greater than that observed among donors. This inference holds even after accounting for project-specific and temporal fixed effects. This finding amplifies the pivotal role of the perceived value surplus and the pursuit of an equitable surplus distribution between funders and project creators in shaping the over-pledging decisions made by funders. Successful fundraising, which fulfills a project's target, can act as an effective quality signal for the product or service, thereby fostering an enhanced perception of fair value.

### 4.2. Imapct of Social Media on OP Score

Social media has emerged as a potent marketing instrument, enabling project creators to engage with an extensive customer base. While it is intuitive that effective social media campaigns could facilitate a wider outreach, the effect of such campaigns on the over-pledging intensity from backers remains ambiguous. We attempt to address this question by capitalizing on a variation in web traffic generated by an exogenous shock. Specifically, Kickstarter introduced a feature known as *Custom Referral Tags* on October 19, 2016.[6] This function empowers creators to monitor incoming viewing traffic from various platforms (e.g., Facebook, Instagram, and Twitter, etc.) through the use of customizable URLs, or referral tags. The outcomes of these referral tags are then displayed on the creator's dashboard, indicating the number of pledges generated, the percentage of total funds accounted for, and the total amount pledged.[7] This newfound ability to understand traffic sources encourages project creators to increase their presence on the most impactful social network channels for driving traffic.

We construct a daily-level panel data with ±40 days surrounding the date when *Custom Referral Tags* was launched.[8] Using the panel data set, we run regressions with the following specification:

$$Y_{it} = \beta_0 + \beta_1 AfterReftag_{it} + ProgressControls + FEs + \epsilon_{it}, \tag{3}$$

where the subscript $i$ indicates the project and $t$ represents the day. We run four regressions with the dependent variable $Y_{it}$ as the overall OP score, OP score by funders, OP score by donors, and the proportion of click-through traffic from social media platforms (including Facebook, Instagram, Twitter and Reddit) to total click-through traffic. The feasibility of the final regression is attributable to the comprehensive nature of our dataset, which includes hourly traffic directed to

[6] See Figure A.1 in the e-companion for the supplement of the introduction date of *Custom Referral Tags*.

[7] See "How do I track referral stats?" (https://help.kickstarter.com/hc/en-us/articles/115005138933-How-do-I-track-referral-stats-)

[8] This represents the maximum time frame we can employ to circumvent the influence of the festive period in December.

each crowdfunding project via various online channels. This data assists in substantiating anecdotal indications that project creators indeed allocate increased resources towards marketing campaigns on social media platforms.

The key independent variable is $AfterReftag_{it}$, indicating whether day $t$ for project $i$ is after the date when *Custom Referral Tags* was introduced on Kickstarter ($=1$ if yes, and $=0$ otherwise). Besides, we also include two control variables (denoted by $ProgressControls$) to account for the relative progress of a project: (i) the number of days since the date when the project was launched; and (ii) whether the funding target was reached by day $t$ for project $i$ ($=1$ of yes, and $=0$ otherwise). Lastly, project fixed effects and launch month fixed effects are also included.

**Table 15 Impact of Custom Referral Tags.**

| | (1) OP Score | (2) OP Score (donors) | (3) OP Score (funders) | (4) Traffic % from Social Media |
|---|---|---|---|---|
| $AfterReftag$ | 0.010*** | 0.009*** | 0.001 | 0.012*** |
| | (0.004) | (0.003) | (0.003) | (0.003) |
| $ProgressControls$ | Yes | Yes | Yes | Yes |
| Project fixed effects | Yes | Yes | Yes | Yes |
| Launch month fixed effects | Yes | Yes | Yes | Yes |
| Observations | 130,486 | 130,486 | 130,486 | 130,486 |
| Adjusted $R^2$ | 0.368 | 0.213 | 0.387 | 0.487 |

*Note*: The dependent variables for columns (1) to (3) are OP scores (overall, funders and donors, respectively); for column(4), it is the proportion of click-through traffic from social media platforms to total click-through traffic. The key independent variable is $AfterReftag_{it}$, which equals 1 if *Custom Referral Tags* was launched on or before day $t$ for project $i$. $ProgressControls$ accounts for all the controls for the relative progress of a project: (i) the number of days since the date when the project was launched; and (ii) whether the funding target was reached by day $t$ for project $i$ ($=1$ of yes, and $=0$ otherwise). Lastly, project fixed effects and launch month fixed effects are also included. $^{***}$ $p<0.01$, $^{**}$ $p<0.05$, $^{*}$ $p<0.1$.

The findings are summarized in Table 15. Firstly, column (4) reinforces anecdotal evidence that, with the ability to trace traffic sources, the proportion of traffic originating from social media markedly increases following the implementation of *Custom Referral Tags*. This can be potentially attributed to intensified activities from the most effective social network channels that are driving the traffic. Intriguingly, a stark dichotomy is observable between the reactions of donors and funders to amplified social media activities. Specifically, the over-pledging intensity of donors increases significantly, while the decision-making of funders seems to remain relatively unaffected,

as demonstrated in columns (2) and (3) respectively. As previously discussed, the over-pledging decisions of donors are potentially driven by altruistic and/or strategic considerations. As indicated in the context of online lending - a somewhat analogous setting - it's evident that such motivations are often influenced by the degree of similarity between lenders and borrowers (Burtch et al. 2014, Lin and Viswanathan 2016). The principle of homophily, a sociological concept highlighting the inclination of individuals to connect with those similar to themselves, is widespread in social networks. Consequently, increased traffic from these networks tends to amplify the over-pledging of donors. However, variations in traffic composition don't impact the perceived value of the products or services at hand, resulting in consistent over-pledging tendencies among funders.

## 5. Discussion and Conclusion

In this paper, we provide a descriptive study of over-pledging behavior in online reward-based crowdfunding. We demonstrate that over-pledging behavior is prevalent in online crowdfunding platforms and has a substantial impact on the project's success rate. We identify two types of over-pledging backers: funders who pay an extra amount for the products in exchange, and donors who pledge the payments without requesting any products. To understand deeper motives of funders' and donors' over-pledging behaviors, we estimate the correlations between over-pledging score (OP score, defined as the proportion of over-pledging, overall or from funders or donors separately, to total pledges for each project) and three aspects of correlates: (i) project- and creator-related characteristics, (ii) funding progress, i.e., before vs after the cumulative pledges reach the target of a (successful) project, and (iii) click-through traffic from online social media platforms.

The findings from our empirical investigation suggest that the amount pledged by donors is driven by the perceived need of a project. We observe an amplified intensity of over-pledging from donors for larger-scale projects, those led by inexperienced creators, and those overseen by creators based in the United States. This observation lends support to the potential motivations for donors' contributions being underpinned by (pure or impure) altruism, alongside other strategic considerations. Conversely, when contrasting the over-pledging patterns of funders with those of donors, we discern that inequity aversion substantially impacts funders' propensity for over-pledging. Leveraging an exogenous shock at the project level, we further ascertain that funders' perceived fair value tends to escalate in response to successful crowdfunding initiatives.

Our results imply that the creators of crowdfunding projects should pay close attention to the over-pledging in order to increase the success rate. Moreover, to increase a project's over-pledging intensity, the creators need to be aware of the different motives and behavior of funders and donors.

For instance, if a creator has designed a good product and expects popular responses from a broad market, the creator may focus on motivating the funders by increasing the perceived fair values of the product. On the other hand, for a creator of a product with social values but expecting a narrow market appeal, the focus should be reaching out to more donors, especially before achieving the funding goal. This can be particularly important for a creator managing the first crowdfunding project. As a caveat about any recommendations based on our results, the analysis is descriptive in nature. Future research may validate the causal inferences through experiments. For example, if a creator of a theatre project alters the funding target or the number of reward levels, would the project attract a different number of donors?

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# E-Companion of "Over-Pledging in Crowdfunding: Evidence from Kickstarter"

## A.1. Summary of Data Filtering

Table A.1 summarizes the steps we followed in filtering out cases from analysis.

**Table A.1 Sample Selection**

| | # of projects filtered-out | # of projects remaining |
|---|---|---|
| (1) Either "successful" or "failed" | 1,985 | 16,426 |
| (2) \$20 $\leq$ Funding target $\leq$ \$500,000 | 164 | 16,262 |
| (3) Duration $\geq$ 48 hours | 5 | 16,257 |
| (4) Backers $\geq$ 5 | 5,038 | 11,219 |
| Selected sample | | 11,219 |

*Note*: *The data set initially consists of 18,411 projects. Criteria (1) to (4) are applied one after another sequentially.*

## A.2. Supplementary Tables for Profiles of Over-Pledging Clusters.

Below we present detailed summary statistics of projects within each of three clusters.

**Table A.2 Summary Statistics of the Top** 1/3

| | Mean | Std. Dev. | Median | Minimum | Maximum |
|---|---|---|---|---|---|
| **Project Characteristics** | | | | | |
| Funding target (\$) | 13,435.24 | 30,218.98 | 5,000 | 25 | 485,000 |
| Duration (hours) | 774.36 | 252.58 | 720 | 117 | 1,538 |
| Number of reward levels | 8.52 | 5.81 | 7 | 1 | 69 |
| Video (= 1 if yes; = 0 otherwise) | 0.71 | 0.45 | 1 | 0 | 1 |
| **Creator Characteristics** | | | | | |
| First time (= 1 if yes; = 0 otherwise) | 0.83 | 0.37 | 1 | 0 | 1 |
| Number of projects backed before | 2.83 | 13.59 | 0 | 0 | 488 |
| Number of collaborators | 0.43 | 0.91 | 0 | 0 | 10 |
| US (= 1 if yes; = 0 otherwise) | 0.71 | 0.46 | 1 | 0 | 1 |
| **Campaign Dynamics** | | | | | |
| Update (# per hour) | 0.01 | 0.01 | 0 | 0 | 0.27 |
| Share (# per hour) | 0.01 | 0.03 | 0.01 | 0 | 0.54 |
| **Outcome** | | | | | |
| Total number of backers | 66.7 | 107.93 | 34 | 6 | 1997 |
| Total pledge amount (\$) | 5,748.35 | 10,080.15 | 2341.48 | 14 | 166,513 |
| Number of OP backers | 20.42 | 35.77 | 11 | 1 | 1,004 |
| Over-pledging (\$) | 2,282.84 | 3,881.93 | 920.12 | 3 | 51,677 |
| Over-pledging/Total pledge amount | 0.43 | 0.19 | 0.38 | 0.21 | 1 |
| Over-pledging from funders/Total pledge amount | 0.14 | 0.16 | 0.08 | 0 | 1 |
| Over-pledging from donors/Total pledge amount | 0.29 | 0.21 | 0.25 | 0 | 1 |

**Table A.3 Summary Statistics of the Middle** 1/3

| | Mean | Std. Dev. | Median | Minimum | Maximum |
|---|---|---|---|---|---|
| **Project Characteristics** | | | | | |
| Funding target ($) | 16,155.71 | 33,509.13 | 6,500 | 20 | 450,000 |
| Duration (hours) | 760.17 | 228.98 | 720 | 72 | 2,093 |
| Number of reward levels | 10.65 | 6.32 | 9 | 1 | 93 |
| Video (= 1 if yes; = 0 otherwise) | 0.73 | 0.44 | 1 | 0 | 1 |
| **Creator Characteristics** | | | | | |
| First time (= 1 if yes; = 0 otherwise) | 0.72 | 0.45 | 1 | 0 | 1 |
| Number of projects backed before | 8.44 | 27.62 | 1 | 0 | 617 |
| Number of collaborators | 0.49 | 0.94 | 0 | 0 | 13 |
| US (= 1 if yes; = 0 otherwise) | 0.67 | 0.47 | 1 | 0 | 1 |
| **Campaign Dynamics** | | | | | |
| Update (# per hour) | 0.01 | 0.02 | 0.01 | 0 | 0.25 |
| Share (# per hour) | 0.03 | 0.08 | 0.01 | 0 | 2.19 |
| **Outcome** | | | | | |
| Total number of backers | 208.2 | 723.36 | 71 | 6 | 24,155 |
| Total pledge amount ($) | 19,548.22 | 170,327.53 | 3,934.99 | 15 | 9,420,134 |
| Number of OP backers | 30.73 | 121.48 | 12 | 1 | 4,047 |
| Over-pledging ($) | 1955.95 | 16277.11 | 431 | 1 | 843,433 |
| Over-pledging/Total pledge amount | 0.12 | 0.05 | 0.11 | 0.05 | 0.21 |
| Over-pledging from funders/Total pledge amount | 0.06 | 0.04 | 0.05 | 0 | 0.21 |
| Over-pledging from donors/Total pledge amount | 0.06 | 0.05 | 0.05 | 0 | 0.21 |

## A.3. Correlation Table.

Table A.5 summarizes pairwise correlations between variables in the regressions in the subsection on primary drivers for over-pledging.

**Table A.4 Summary Statistics of the Bottom** 1/3

| | Mean | Std. Dev. | Median | Minimum | Maximum |
|---|---|---|---|---|---|
| **Project Characteristics** | | | | | |
| Funding target ($) | 17889.31 | 33832.32 | 7326.03 | 20 | 497805.68 |
| Duration (hours) | 777.79 | 255.79 | 720 | 72 | 1440 |
| Number of reward levels | 9.87 | 6.7 | 9 | 1 | 179 |
| Video (= 1 if yes; = 0 otherwise) | 0.7 | 0.46 | 1 | 0 | 1 |
| **Creator Characteristics** | | | | | |
| First time (= 1 if yes; = 0 otherwise) | 0.63 | 0.48 | 1 | 0 | 1 |
| Number of projects backed before | 8.98 | 29.87 | 1 | 0 | 749 |
| Number of collaborators | 0.42 | 0.98 | 0 | 0 | 10 |
| US (= 1 if yes; = 0 otherwise) | 0.61 | 0.49 | 1 | 0 | 1 |
| **Campaign Dynamics** | | | | | |
| Update (# per hour) | 0.01 | 0.01 | 0.01 | 0 | 0.18 |
| Share (# per hour) | 0.03 | 0.21 | 0.01 | 0 | 8.22 |
| **Outcome** | | | | | |
| Total number of backers | 464.13 | 3268.06 | 65 | 6 | 158514 |
| Total pledge amount ($) | 35941.26 | 186956.87 | 3661.74 | 18 | 6432418.51 |
| Number of OP backers | 23.46 | 103.29 | 5 | 0 | 3577 |
| Over-pledging ($) | 845.92 | 4421.12 | 67 | 0 | 167116.09 |
| Over-pledging/Total pledge amount | 0.02 | 0.02 | 0.02 | 0 | 0.05 |
| Over-pledging from funders/Total pledge amount | 0.01 | 0.01 | 0.01 | 0 | 0.05 |
| Over-pledging from donors/Total pledge amount | 0.01 | 0.01 | 0 | 0 | 0.05 |

**Table A.5 Pairwise Correlations.**

| | OP score | log(target) | reward | video | first time | backed | collab | US | update | share |
|---|---|---|---|---|---|---|---|---|---|---|
| OP score | 1 | | | | | | | | | |
| log(target) | -0.06*** | 1 | | | | | | | | |
| reward | -0.16*** | 0.23*** | 1 | | | | | | | |
| video | -0.02** | 0.19*** | 0.09*** | 1 | | | | | | |
| first time | 0.17*** | 0.18*** | -0.05*** | 0.03*** | 1 | | | | | |
| backed | -0.11*** | -0.07*** | 0.05*** | 0.00 | -0.28*** | 1 | | | | |
| collab | -0.02** | 0.19*** | 0.12*** | 0.08*** | 0.11*** | 0.01 | 1 | | | |
| US | 0.06*** | 0.02* | 0.04*** | 0.01 | -0.10*** | 0.09*** | 0.01 | 1 | | |
| update | -0.19*** | 0.00 | 0.17*** | 0.01*** | -0.24*** | 0.21*** | 0.09*** | 0.03** | 1 | |
| share | -0.06*** | 0.12*** | 0.08*** | 0.04*** | -0.06*** | 0.12*** | 0.08*** | 0.03** | 0.15*** | 1 |

*Note:* $^{***}\ p < 0.01$, $^{**}\ p < 0.05$, $^{*}\ p < 0.1$.

## A.4. Supplements about Custom Referral Tags.

**Figure A.1 Introduction Date of Custom Referral Tags on Kickstarter**

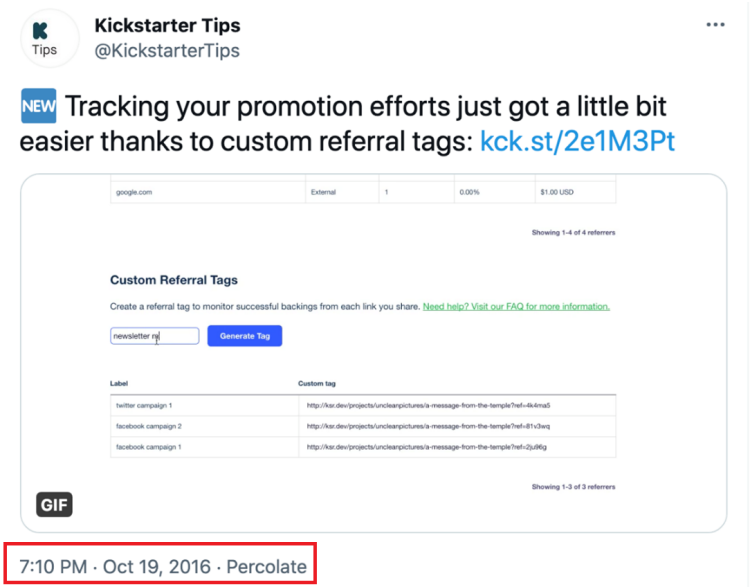
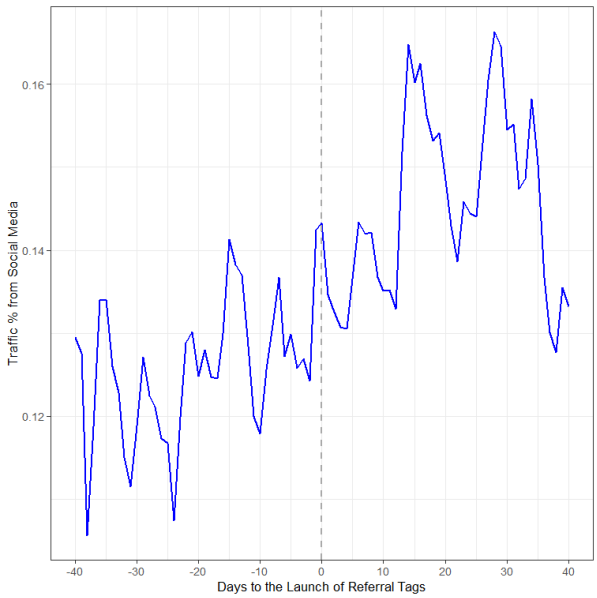


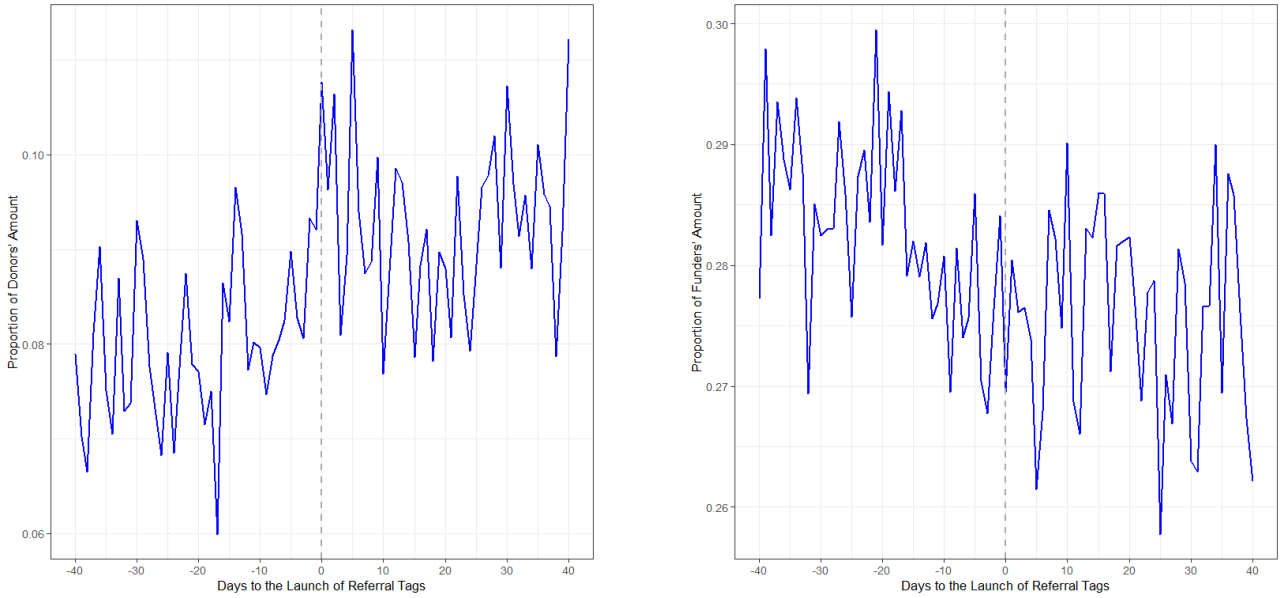

**Figure A.2 Proportion of Donors' Contribution Among All Backers.**




(a) Count

*Note.* The x-axis indicates the number of days to Oct. 19, 2016, which is the date when the function *Referral Tags* was launched. The y-axis denotes the average proportion of traffic from personal social media platforms among all the click-through traffic.

**Figure A.3 Proportion of Donors' Contribution Among All Backers.**




(a) Donors (b) Funders

*Note.* The x-axis indicates the number of days to Oct. 19, 2016, which is the date when the function *Custom Referral Tags* was launched. The y-axis denotes the over-pledging score for donors (a) or for funders (b).